Xutong Liu

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Research Interests

My research focuses on developing **structure-aware online learning and reinforcement learning (RL)** algorithms that leverage inherent **action, feedback, and agent structures**—such as smoothness, sparsity, and clustering—to enable **data-efficient, scalable, and robust** decision-making in networked systems. My goal is to bridge theory and real-world applications for LLM **inference systems, edge/cloud computing, and conversational recommender systems**.

Professional Experience

University of Washington - Tacoma — **United States**
Tenure-Track Assistant Professor, Department of Computer Science and Systems — *Sept 2025 – Present*

Carnegie Mellon University — **United States**
Postdoc Research Fellow, Department of Electrical and Computer Engineering — *Sept 2024 – Sept 2025*
Advisor: Prof. Carlee Joe-Wong

University of Massachusetts Amherst — **United States**
Visiting Postdoc Researcher, Manning College of Information and Computer Sciences — *Oct 2023 – Aug 2024*
Advisor: Prof. Mohammad Hajiesmaili

The Chinese University of Hong Kong — **Hong Kong SAR**
RGC Postdoc Research Fellow, Department of Computer Science and Engineering — *Sept 2022 – Aug 2024*
Advisor: Prof. John C.S. Lui

Microsoft Research Asia — **China**
Research Intern, Theory Group (now Theory Center) — *June 2019–Sept 2019*
Mentor: Dr. Wei Chen

Education

The Chinese University of Hong Kong — **Hong Kong SAR**
Ph.D., Department of Computer Science and Engineering — *Sept 2017–July 2022*
Thesis: Efficient Online Learning Algorithms for Combinatorial Optimization Problems: Framework, Theory and Applications
Advisor: John C.S. Lui

University of Science and Technology of China — **China**
B.Eng., School of Computer Science and Technology — *Sept 2013–June 2017*

Honors and Awards

Best Paper Finalists (Top 5), ACM SIGMETRICS	*2025*
RGC Postdoctoral Fellowship (50 recipients worldwide, support 1.2M HK dollars for three years)	*2023*
Award of Excellence in Stars of Tomorrow Internship Program, Microsoft Research Asia	*2019*
Hong Kong PhD Fellowship (top 5%)	*2017*
Runner up, Guo Moruo Scholarship, USTC (the highest honor)	*2017*
Bachelor's Degree with Honored Rank, USTC (top 3%)	*2017*
Elite Collegiate Award, China Computer Federation	*2016*
National Scholarship, Ministry of Education of China (top 0.2%)	*2014, 2015*

Selected Publications

[1] **Xutong Liu**, Xiangxiang Dai, Xuchuang Wang, Mohammad Hajiesmaili, John C.S. Lui. Combinatorial Logistic Bandits. *ACM **SIGMETRICS*** (**Best Paper Runner-Up**), 2025.

[2] **Xutong Liu**, Jinhang Zuo, Xiaowei Chen, Wei Chen, John C.S. Lui. Multi-layered Network Exploration via Random Walks: From Offline Optimization to Online Learning. *Thirty-eighth International Conference on Machine Learning (**ICML**)*, pp. 7057-7066, 2021. (**Long oral, top 3% from 5513 submissions**)

[3] **Xutong Liu**, Jinhang Zuo, Siwei Wang, Carlee Joe-Wong, John C.S. Lui, Wei Chen. Batch-Size Independent Regret Bounds for Combinatorial Semi-Bandits with Probabilistically Triggered Arms or Independent Arms. *Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS**)*, pp.14904-14916 2022.

Full Publications

Conference Publications

[1] Mengfan Xu, Liren Shan, Fatemeh Ghaffari, Xuchuang Wang, **Xutong Liu**, and Mohammad Hajiesmaili. Heterogeneous Multi-agent Multi-armed Bandits on Stochastic Block Models. Accepted by ACM International Conference on Measurement and Modeling of Computer Systems (**SIGMETRICS**), 2026 (23/111 = 20.7%).

[2] Qijia He, Minghan Wang, **Xutong Liu**, Zhiyong Wang, Fang Kong. Learning Across the Gap: Hybrid Multi-armed Bandits with Heterogeneous Offline and Online Data. Accepted by the Thirty-nineth Conference on Neural Information Processing Systems (**NeurIPS**), 2025. (5290/21575=24.5%).

[3] Xiangxiang Dai, Xiaowei Sun, Jinhang Zuo, **Xutong Liu**#, John C.S. Lui. Offline Learning for Combinatorial Multi-armed Bandits. ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**) 2025. (Acceptance rate: 18.4%)

[4] **Xutong Liu**, Xiangxiang Dai, Jinhang Zuo, Siwei Wang, Xuchuang Wang, Carlee Joe-Wong, John C.S. Lui, and Wei Chen. Offline Learning for Combinatorial Multi-armed Bandits. Forty-second International Conference on Machine Learning (**ICML**), 2025. (Acceptance rate: 26.3%)

[5] Xuchuang Wang, Qirun Zeng, Jinhang Zuo, **Xutong Liu**#, Mohammad Hajiesmaili, John C.S. Lui, Adam Wierman. Fusing Reward and Dueling Feedback in Stochastic Bandits. Forty-second International Conference on Machine Learning (**ICML**), 2025. (Acceptance rate: 26.3%)

[6] **Xutong Liu**, Xiangxiang Dai, Xuchuang Wang, Mohammad Hajiesmaili, John C.S. Lui. Combinatorial Logistic Bandits. *ACM **SIGMETRICS*** (**Best Paper Runner-Up**), 2025. (Acceptance rate: 18.2%)

[7] **Xutong Liu**, Jinhang Zuo, Junkai Wang, Zhiyong Wang, Yuedong Xu, John C.S. Lui. Learning Context-Aware Probabilistic Maximum Coverage Bandits: A Variance-Adaptive Approach. *IEEE Conference on Computer Communications (**INFOCOM**)*, pp. 2189-2198, 2024. (Acceptance rate: 19.6%)

[8] **Xutong Liu***, Jinhang Zuo*, Hong Xie, Carlee Joe-Wong, John C.S. Lui. Variance-Adaptive Algorithm for Probabilistic Maximum Coverage Bandits with General Feedback. *IEEE International Conference on Computer Communications (**INFOCOM**)*, pp. 1-10, 2023. (*Equal contribution. Acceptance rate: 19.2%)

[9] **Xutong Liu**, Jinhang Zuo, Xiaowei Chen, Wei Chen, John C.S. Lui. Multi-layered Network Exploration via Random Walks: From Offline Optimization to Online Learning. *Thirty-eighth International Conference on Machine Learning (**ICML**)*, pp. 7057-7066, 2021. (**Long oral (3%)**. Acceptance rate: 21.5%)

[10] **Xutong Liu**, Jinhang Zuo, Siwei Wang, Carlee Joe-Wong, John C.S. Lui, Wei Chen. Batch-Size Independent Regret Bounds for Combinatorial Semi-Bandits with Probabilistically Triggered Arms or Independent Arms. *Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS**)*, pp.14904-14916 2022. (Acceptance rate: 25.5%)

[11] **Xutong Liu**, Siwei Wang, Jinhang Zuo, Han Zhong, Xuchuang Wang, Zhiyong Wang, Shuai Li, Mohammad Hajiesmaili, John C.S. Lui, Wei Chen. Combinatorial Multivariant Multi-Armed Bandits with Applications to Episodic Reinforcement Learning and Beyond. *Forty-first International Conference on Machine Learning (**ICML**)*, 2024. (Acceptance rate: 27.5%).

[12] **Xutong Liu**, Jinhang Zuo, Siwei Wang, John C.S. Lui, Mohammad Hajiesmaili, Adam Wierman, Wei Chen. Contextual Combinatorial Bandits with Probabilistically Triggered Arms. *The Fortieth International Conference on Machine Learning (**ICML**)*, pp. 22559-22593, 2023. (Acceptance rate: 27.9%)

[13] **Xutong Liu**, Haoru Zhao, Tong Yu, Shuai Li, John C.S. Lui. Federated Online Clustering of Bandits. *The 38th Conference on Uncertainty in Artificial Intelligence (**UAI**)*, pp. 1221-1231 2022. (Acceptance rate: 32.0%)

[14] Xuchuang Wang, Maoli Liu#, **Xutong Liu**#, Zhuohua Li, Mohammad Hajiesmaili, John C.S. Lui, Don Towsley. Learning Best Paths in Quantum Networks. *IEEE Conference on Computer Communications (**INFOCOM**)*, 2025. (#Corresponding author. Acceptance rate: 18.7%)

[15] Junkai Wang, **Xutong Liu**#, Jinhang Zuo, Yuedong Xu#. Robust Contextual Combinatorial Multi-Armed Bandits for Unreliable Network Systems. *IEEE Conference on Computer Communications (**INFOCOM**)*, 2025. (#Corresponding author. Acceptance rate: 18.7%)

[16] Xiangxiang Dai, Zeyu Zhang, Peng Yang, Yuedong Xu, **Xutong Liu**#, John C.S. Lui. AxiomVision: Accuracy-Guaranteed Adaptive Visual Model Selection for Perspective-Aware Video Analytics. *Thirty-second ACM Multimedia Conference. (**MM**)*, 2024. (#Corresponding author. Acceptance rate: 26.2%)

[17] Kechao Cai, **Xutong Liu**, Yuzhen Janice Chen, and John C.S. Lui. An Online Learning Approach to Network Application Optimization with Guarantee. *IEEE International Conference on Computer Communications (**INFOCOM**)*, pp. 2006-2014, 2018. (Acceptance rate: 19.2%)

[18] Xuchuang Wang, Maoli Liu, Jinhang Zuo, **Xutong Liu**#, John C.S. Lui, and Mohammad Hajiesmaili. Stochastic Bandits Robust to Adversarial Attacks. *The Thirteenth International Conference on Learning Representations (**ICLR**)*, 2025. (#Corresponding author. Acceptance rate: 32.1%)

[19] Hantao Yang, **Xutong Liu**#, Zhiyong Wang, Hong Xie, John C.S. Lui, Defu Lian, and Enhong Chen. Federated Contextual Cascading Bandits with Asynchronous Communication and Heterogeneous Users. *The 38th AAAI Conference on Artificial Intelligence (**AAAI**)*, vol. 38, no. 18, pp. 20596-20603, 2024. (#Corresponding author. Acceptance rate: 23.75%)

[20] Xuchuang Wang, Yu-Zhen Janice Chen, Lin Yang, **Xutong Liu**, Mohammad Hajiesmaili, Don Towsley, and John C.S. Lui. Asynchronous Multi-Agent Bandits: Fully Distributed vs. Leader-Coordinated Algorithms. *ACM **SIGMETRICS***, 2025. (Acceptance rate: 18.2%)

[21] Jianhao He, Chengchang Liu, **Xutong Liu**, Lvzhou Li, John C.S. Lui. Quantum Algorithm for Online Exp-concave Optimization. *Forty-first International Conference on Machine Learning (**ICML**)*, 2024. (Acceptance rate: 27.5%).

[22] Zhiyong Wang, Jize Xie, **Xutong Liu**, Shuai Li, John C.S. Lui. Online Clustering of Bandits with Misspecified User Models. *Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS**)*, 2023. (Acceptance rate: 26.1%)

[23] Xuchuang Wang, Lin Yang, Yu-zhen Janice Chen, **Xutong Liu**, Mohammad Hajiesmaili, Don Towsley, John C.S. Lui. Exploration for Free: How Does Reward Heterogeneity Improve Regret in Cooperative Multi-agent Bandits? *Thirty-ninth Conference on Uncertainty in Artificial Intelligence (**UAI**)*, pp. 2192-2202, 2023. (Acceptance rate: 31.2%)

[24] Yu-Zhen Janice Chen, Lin Yang, Xuchuang Wang, **Xutong Liu**, Mohammad Hajiesmaili, John C.S. Lui, Don Towsley. On-Demand Communication for Asynchronous Multi-Agent Bandits. *The 26th International*

*Conference on Artificial Intelligence and Statistics (**AISTATS**)*, pp. 3903-3930, 2023. (Acceptance rate: 29.4%)

[25] Xuchuang Wang, Lin Yang, Yu-Zhen Janice Chen, **Xutong Liu**, Mohammad Hajiesmaili, Don Towsley, John C.S. Lui. Achieve Near-Optimal Individual Regret and Low Communications in Multi-Agent Bandits. *The Eleventh International Conference on Learning Representations (**ICLR**)*, 2023. (Acceptance rate: 32.0%)

[26] Jinhang Zuo, **Xutong Liu**, Carlee Joe-Wong, John C.S. Lui, Wei Chen. Online Competitive Influence Maximization. *International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, pp. 11472-11502, 2022. (Acceptance rate: 29.1%)

[27] Zhiyong Wang, **Xutong Liu**, Shuai Li, John C.S. Lui. Efficient Explorative Key-term Selection Strategies for Conversational Contextual Bandits. *Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI**)*, vol. 37, no. 8, pp. 10288-10295, 2023. (Acceptance rate: 19.6%)

Journal Publications

[1] Anupama Sitaraman, Adam Lechowicz, Noman Bashir, **Xutong Liu**, Mohammad Hajiesmaili, Prashant Shenoy. Dynamic Incentive Allocation for City-Scale Deep Decarbonization. ACM Journal on Computing and Sustainable Societies (**JCSS**), 2025.

[2] Xiangxiang Dai, **Xutong Liu**#, Jinhang Zuo, Hong Xie, Carlee Joe-Wong, John C.S. Lui. Variance-Aware Bandit Framework for Dynamic Probabilistic Maximum Coverage Problem with Triggered or Self-Reliant Arms. IEEE/ACM Transactions on Networking (**TON**), 2024. (#Corresponding author.)

[3] Kechao Cai, **Xutong Liu**, Yuzhen Janice Chen, and John C.S. Lui. Learning with Guarantee via Constrained Multi-armed Bandit: Theory and Network Applications. *IEEE Transactions on Mobile Computing (**TMC**)*, vol. 22, no. 9, pp. 5346-5358, 2022.

[4] Xiangxiang Dai, Zhiyong Wang, Jize Xie, **Xutong Liu**, John C.S. Lui. Conversational Recommendation with Online Learning and Clustering on Misspecified Users. IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2024.

[5] **Xutong Liu**, Yu-Zhen Chen, John C.S. Lui., Konstantin Avrachenkov.. Learning to Count: a Deep Learning Framework for Graphlet Count Estimation. *Network Science*, vol. 9, no. S1, pp. S23-S60, 2021.

Paper in Submission

[1] **Xutong Liu**, Baran Atalar, Xiangxiang Dai, Jinhang Zuo, Siwei Wang, John C.S. Lui, Wei Chen, Carlee Joe-Wong. Semantic Caching for Low-Cost LLM Serving: From Offline Learning to Online Adaptation.

[2] Jingyuan Liu, Zeyu Zhang, Xuchuang Wang, **Xutong Liu**, John C.S. Lui, Mohammad Hajiesmaili, Carlee Joe-Wong. Offline Clustering of Linear Bandits: Unlocking the Power of Clusters in Data-Limited Environments. 2025.

[3] Xiangxiang Dai, Jin Li, **Xutong Liu**, Anqi Yu, John C.S. Lui. Cost-Effective Online Multi-LLM Selection with Versatile Reward Models. 2024.

Academic Service

Publicity Chair: International European Conference on Parallel and Distributed Computing (Euro-Par) 2026
Co-organizer: The 3rd Annual Workshop on Learning-augmented Algorithms: Theory and Applications at ACM SIGMETRICS 2025
Journal Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE/ACM Transactions on Networking, IEEE Transactions on Mobile Computing, IEEE Transactions on Network Science and Engineering
TPC Member: ACM SIGMETRICS, ACM e-Energy
PC Member/Conference Reviewer: NeurIPS, ICML, ICLR, ACM MM, AISTATS, UAI, AAAI

Presentations

Scalable and Robust Online Learning for Complex Networked Systems
– SQUALL at Carnegie Mellon University, Pittsburgh, USA *June 2025*

Combinatorial Logistic Bandits
– ACM SIGMETRICS, Stony Brook, New York, USA *June 2025*

Structure-Aware RL for Complex Networked Systems
– ACM SIGMETRICS Mentoring and Networking Workshop, Stony Brook, New York, USA *June 2025*

Scalable and Robust Online Learning for AI-Powered Networked Systems
– Computer Science and System Department, University of Washington, Tacoma *Feb 2025*

Learning Context-Aware Probabilistic Maximum Coverage Bandits
– IEEE International Conference on Computer Communications (INFOCOM) *May 2024*

A combinatorial Multi-Armed Bandit View to Solve Episodic RL
– John Hopcroft Center, Shanghai Jiao Tong University *April 2024*

Theory and Applications for Multi-agent Online Combinatorial Decision Making
– School of Computer Science and Technology, University of Science and Technology of China *Dec 2023*

Variance-Adaptive Algorithms for Probabilistic Maximum Coverage Bandits
– School of Computer Science and Engineering, Southeast University *April 2023*

Multi-layered Network Exploration via Random Walks: From Offline Optimization to Online Learning
– Thirty-eighth International Conference on Machine Learning (ICML) *July 2021*

Teaching Experience

TCSS142: Programming Principles, University of Wahington - Tacoma
Instructor *Fall 2025*

CSCI2040: Introduction to Python, CUHK
Lead Teaching Assistant *Spring 2019*

CSCI3320: Fundamentals of Machine Learning, CUHK
Lead Teaching Assistant *Fall 2018*

CSCI2040: Introduction to Python, CUHK
Teaching Assistant *Spring 2017,2018*

CSCI3320: Fundamentals of Machine Learning, CUHK
Teaching Assistant *Fall 2017*

Mentorship

Carnegie Mellon University (with Prof. Carlee Joe-Wong), ECE
Baran Atalar (Ph.D. student in ECE at CMU), 2025-Present
Jingyuan Liu (visiting undergraduate student), 2024-Present
Hanqing Yang (master student $\rightarrow$ Ph.D. in ECE at CMU), 2024-Present
UMass Amherst (with Prof. Mohammad Hajiesmaili), CICS
Fatemeh Ghaffari (Ph.D. in CICS at UMass Amherst), 2024-Present: JCSS '25$\times 1$
Anupama Sitaraman (undergraduate student $\rightarrow$ Ph.D. in CS at CMU), 2023-2024
The Chinese University of Hong Kong (with Prof. John C.S. Lui), CSE
Ziyi Han (Ph.D. in CSE at CUHK), 2024-Present
Zeyu Zhang (Ph.D. in CSE at CUHK), 2024-Present
Xiangxiang Dai (Ph.D. in CSE at CUHK), 2023-Present: IEEE/ACM TON $\times 1$, ACM MM' 24 $\times 1$
Zhiyong Wang (Ph.D. in CSE at CUHK), 2022-2023: IEEE TKDE $\times 1$, NeurIPS' 23 $\times 1$, AAAI' 23 $\times 1$
Yu-Zhen Janice CHEN (undergraduate student $\rightarrow$ Ph.D. in CICS at UMass Amherst), 2017-2018: Network Science Journal $\times 1$
Southern University of Science and Technology (with Prof. Fang Kong), SDS

Qijia He (undergraduate student at SUSTech → Ph.D. in ECE at OSU), 2024-2025: NeurIPS '25 $\times 1$.
University of Science and Technology of China (with Prof. Hong Xie and Prof. Defu Lian), CS
Hantao Yang (master student in CS at USTC → Ph.D. in CS at USTC), 2023-Present: AAAI '24 $\times 1$
Fudan University (with Prof. Yuedong Xu), EE
Zhaoying He (master student in EE at FDU), 2025-Present
Junkai Wang (master student in EE at FDU), 2023-Present: INFOCOM' 25 $\times 1$

References

John C.S. Lui
Choh-Ming Li Chair Professor of Computer Science and Engineering
Fellow of ACM, Fellow of IEEE, Croucher Senior Research Fellow
Fellow of the Hong Kong Academy of Engineering (HKAE)
Department of Computer Science and Engineering
The Chinese University of Hong Kong

Carlee Joe-Wong
Robert E. Doherty Professor of Electrical and Computer Engineering
Department of Electrical and Computer Engineering
Carnegie Mellon University

Mohammad Hajiesmaili
Associate Professor at the Manning College of Information and Computer Sciences
Manning College of Information and Computer Sciences
University of Massachusetts Amherst

Wei Chen
Principal Researcher at Microsoft Research
Fellow of ACM, Fellow of IEEE
Chair of MSR Asia Theory Center
Microsoft Research